Building Knowledge Graph for AI Policy Documents

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Key Contributions

Structured Obligations
Identified who is responsible for what in AI-related laws (AGORA legal docs)

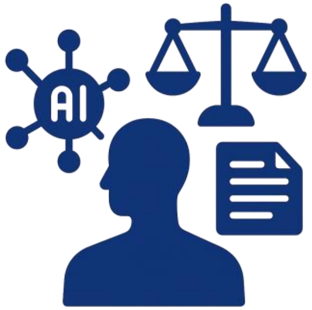
Risk and Technology Mapping
Linked laws to concerns like privacy, safety, and emerging technologies

Query-able Legal Graph
Built a graph in Neo4j that supports questions about oversight and regulation

Agenda

1. Data Analysis
2. Assumptions
3. Methodology
4. Knowledge Graph
5. Graph RAG
6. Limitations & Future Directions

Dataset - AGORA (AI GOvernance and Regulatory Archive)



AGORA Dataset
Source: Zenodo.17621534 (2025)

- A collection of U.S. AI-related laws, bills, executive orders, regulations, and government actions.
- Includes document text, metadata, summaries, and thematic tags.
- Provided by Emerging Technology Observatory.

Dataset - AGORA (AI GOvernance and Regulatory Archive)

Example Case

California Assembly Bill 3030 (2024)

Casual Name: California AB 3030 (AI Health Care Services)

Proposed Date: 2024/02/16

Link: https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240AB3030

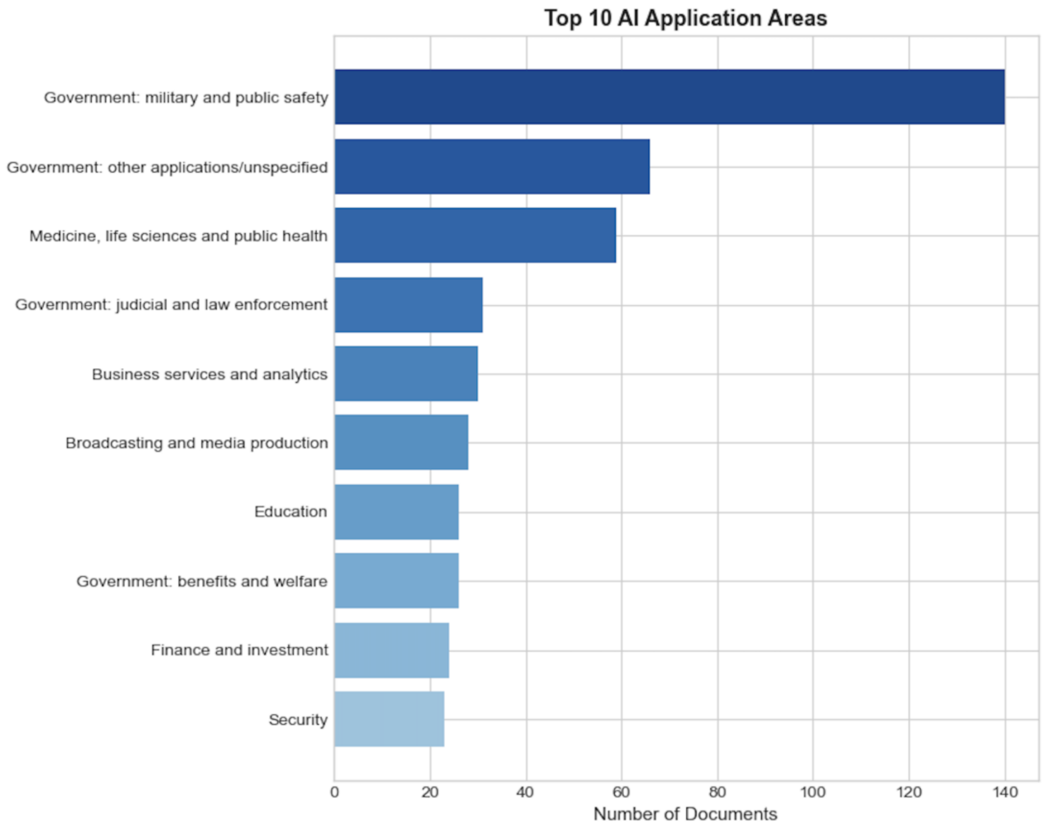
Tags:

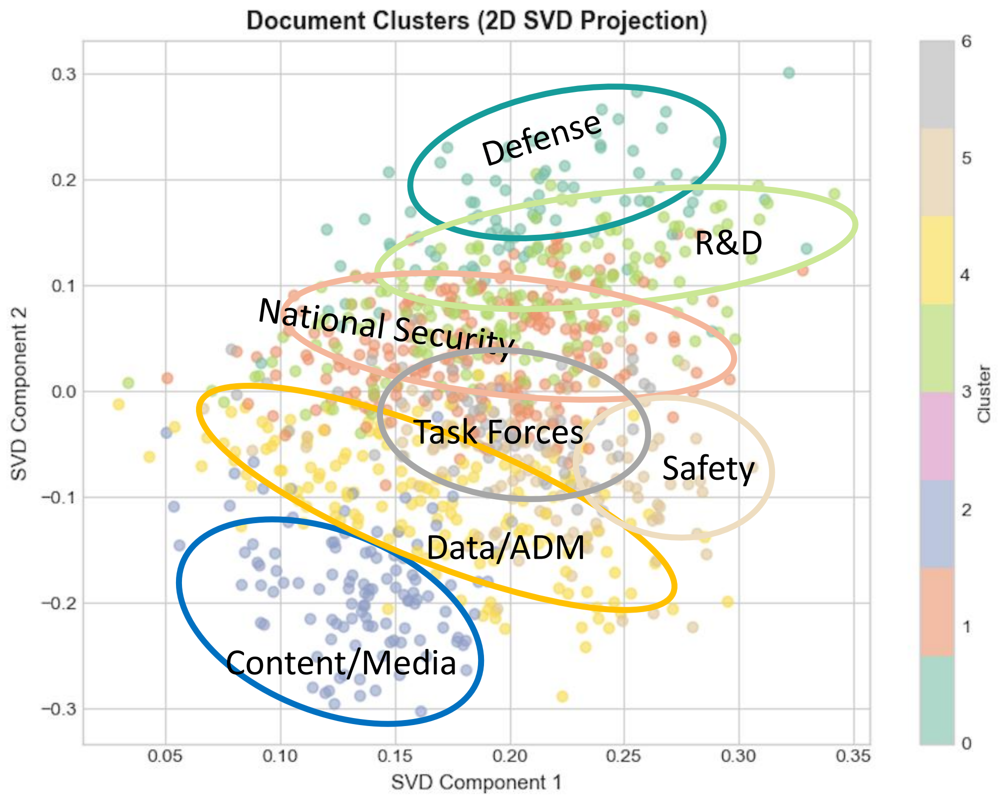
- Risk factors: Transparency;
- Strategies: Disclosure;
- Applications: Medicine, life sciences and public health

Long Summary

- Requires health facilities, clinics, and physician offices using generative AI for patient communications in a variety of modalities to include a disclaimer indicating AI generation.
- Mandates the disclaimer's prominence at the beginning, end, and/or throughout written, audio, and video communications, with instructions for contacting a human provider.
- Exempts communications reviewed by a licensed health care provider from disclaimer requirements.
- Defines key terms, including "artificial intelligence," "generative artificial intelligence," and "patient clinical information."
- Subjects licensed health facilities, clinics, and California Assembly Bill 3030 (2024) physicians violating these requirements to enforcement mechanisms.

National Security, AI Safety and Risks Are Main Regulatory Areas

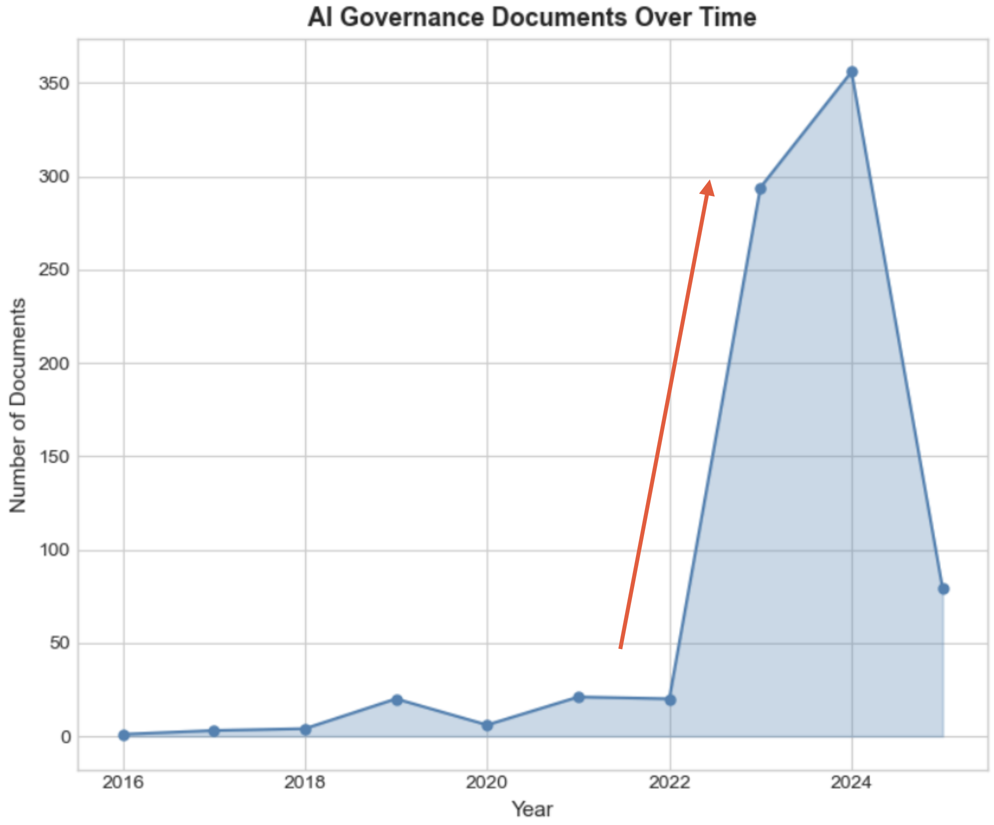


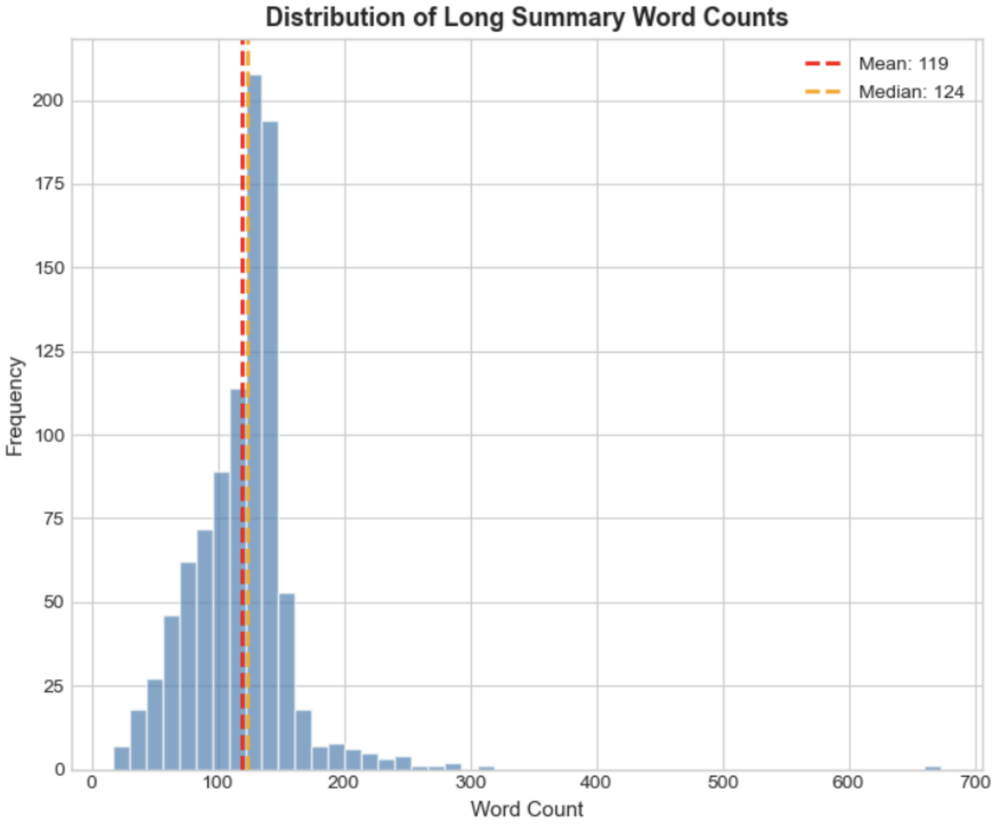


- Defense & public safety dominate (140 docs) — high-risk domains
- Healthcare, law enforcement also heavily regulated

- The thematic clusters identified via K-means on TF-IDF vectors
- 7 clusters identified: Defense (0), National Security (1), Content/Media (2), R&D (3), Data/ADM (4), Safety (5), Task Forces (6)
- Overlap = shared legal language; clusters = thematic emphasis

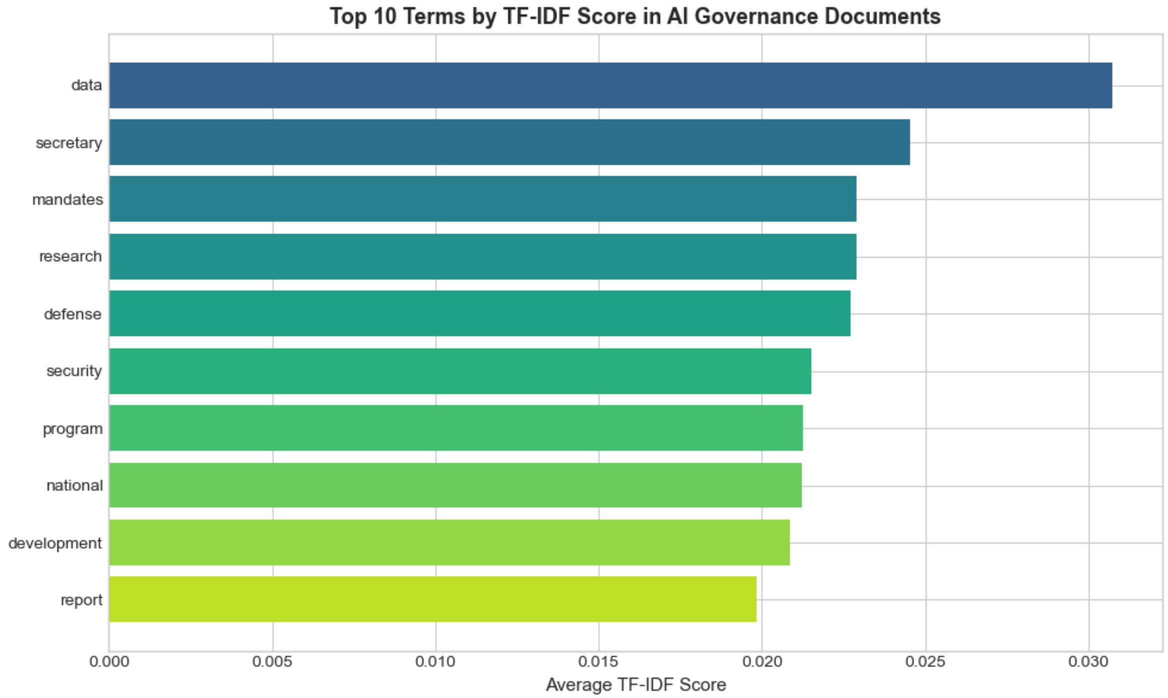
Growth of Documents Reflect the Upward Trend in AI Technology





Number of documents: **951** | Avg words per doc in long summary: **119** **|** Avg tokens per sentence: **23**

Data, Security and Regulation are Key Focus Areas



1. **Data** — Ranks highest, implying data governance is central to AI policy.
2. **Secretary, mandates, defense** — Reflects executive authority and security focus.
3. **Development, report, public** — Reflects the prescriptive, program-building nature of AI governance.

Assumptions

Summaries are good approximation to original text

Entity accuracy = meaningful extraction

Verbs = relationships

Triples are sufficient to represent legal meaning

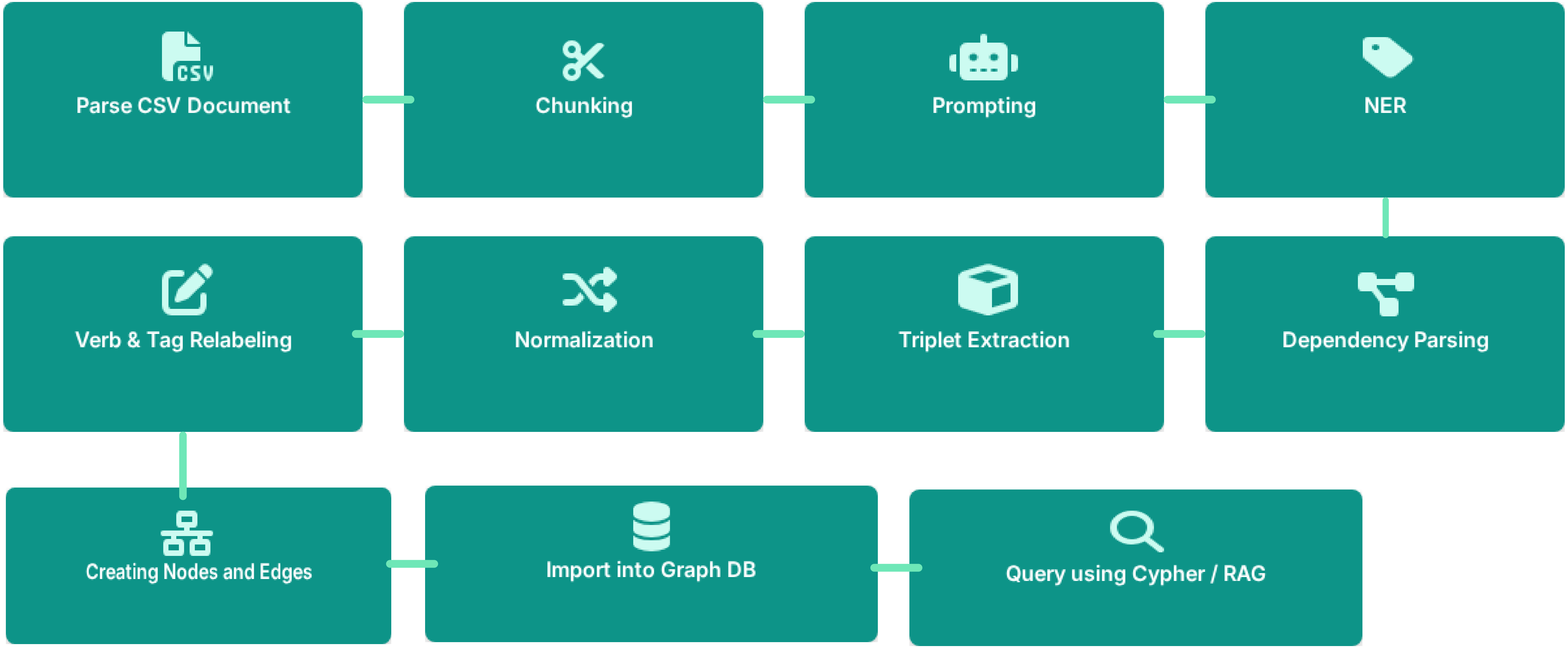
Shallow extraction is okay if you extract enough of it

If a relationship is syntactically captured, it is semantically meaningful

LLMs and spaCy extract structure directly usable for KG

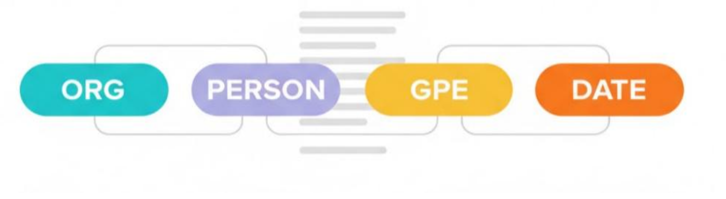
If the KG is built, I can query it for insight

A Multi-Step Pipeline To Build Legal Knowledge Graph



Domain-specific rules turn generic NER into precise legal entities.

Default Extraction



California GPE Executive Order N-12-23 PERSON orders California GPE 's Government Operations Agency ORG and related entities to draft a report for the Governor within 60 days DATE on beneficial uses and risks of Generative AI (GenAI) by the state.

Custom Extraction

{"pattern": **"California Executive Order N-12-23"**,
"label": **"LEGAL_SOURCE"** },
{"pattern": **"California's Government Operations Agency"**,
"label": **"ACTOR"**},
{"pattern": **"report"**,
"label": **"LEGAL_SOURCE“**},
…

California Executive Order N-12-23 LEGAL_SOURCE orders California's Government Operations Agency ACTOR and related entities to draft a report LEGAL_SOURCE for the Governor ACTOR within 60 days DATE on beneficial uses and risks of Generative AI TECHNOLOGY (GenAI TECHNOLOGY) by the state.

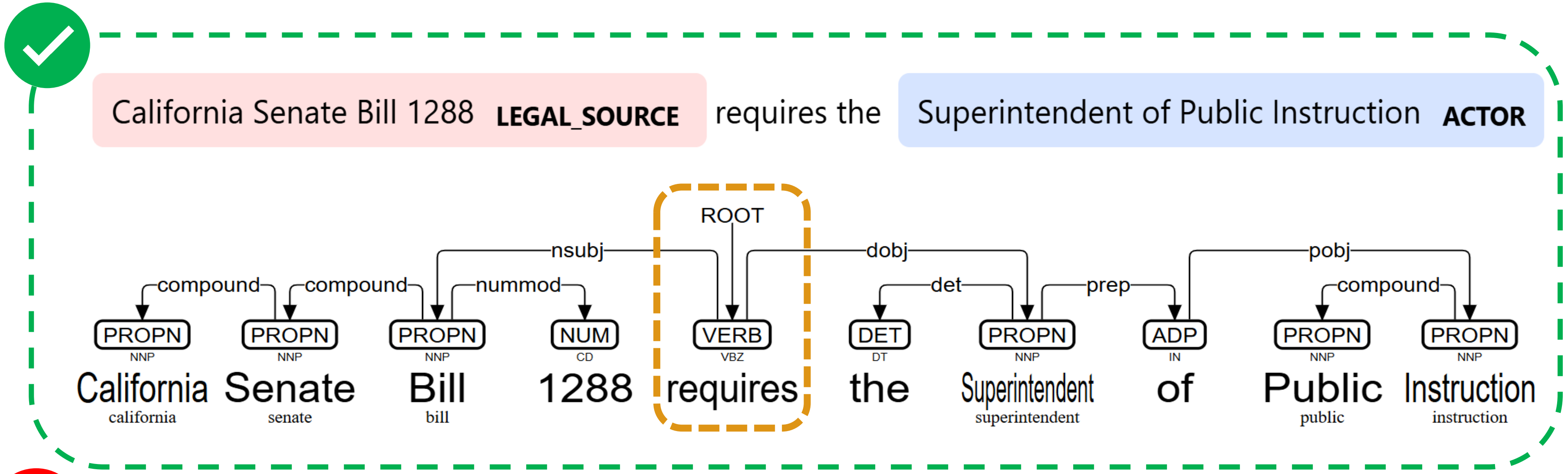
Dependency parsing fails when legal terms double as verbs

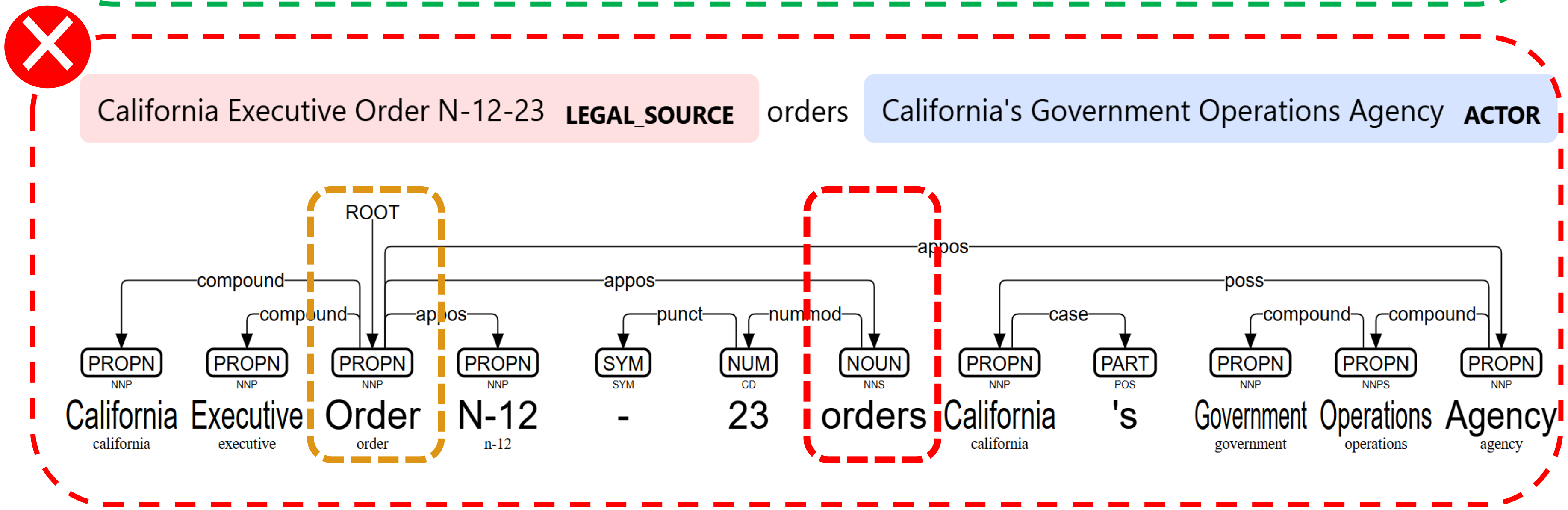
Walking Algorithm

walk(root)

→ find subject (LEGAL_SOURCE)
→ find object (ACTOR)
→ record (LEGAL_SOURCE → action → ACTOR)

"Failure isn't isolated — verbs that double as legal nouns ('order', 'bill', 'report') often break dependency-based extraction."





Extraction captures structure, but loses legal meaning and introduces noise.

California Executive Order N-12-23 | requires | California Cybersecurity Integration Center

Verb Remapping

requires: OBLIGATE

TEXT: "California Executive Order N-12-23 requires the California Cybersecurity Integration Center…"

✅ Clean, strong relationship

California Executive Order N-12-23 | emphasize | collaboration

Verb Remapping

emphasize: OTHER

TEXT: "California Executive Order N-12-23 emphasize collaboration with society, academia and industry experts"

✅ Structurally valid
❌ Semantically shallow

California Executive Order N-12-23 | perform | a risk analysis

Verb Remapping

perform: SUPPORT

TEXT: "California Executive Order N-12-23 perform a risk analysis by March 2024 on GenAI threats to energy…."

❌ Relationship broken over clauses

Hence, we filtered the graph to retain only a small, meaningful set of collapsed legal action types: OBLIGATE, REGULATE, REPORT, etc.

Structure of the Extracted Knowledge Graph

Nodes

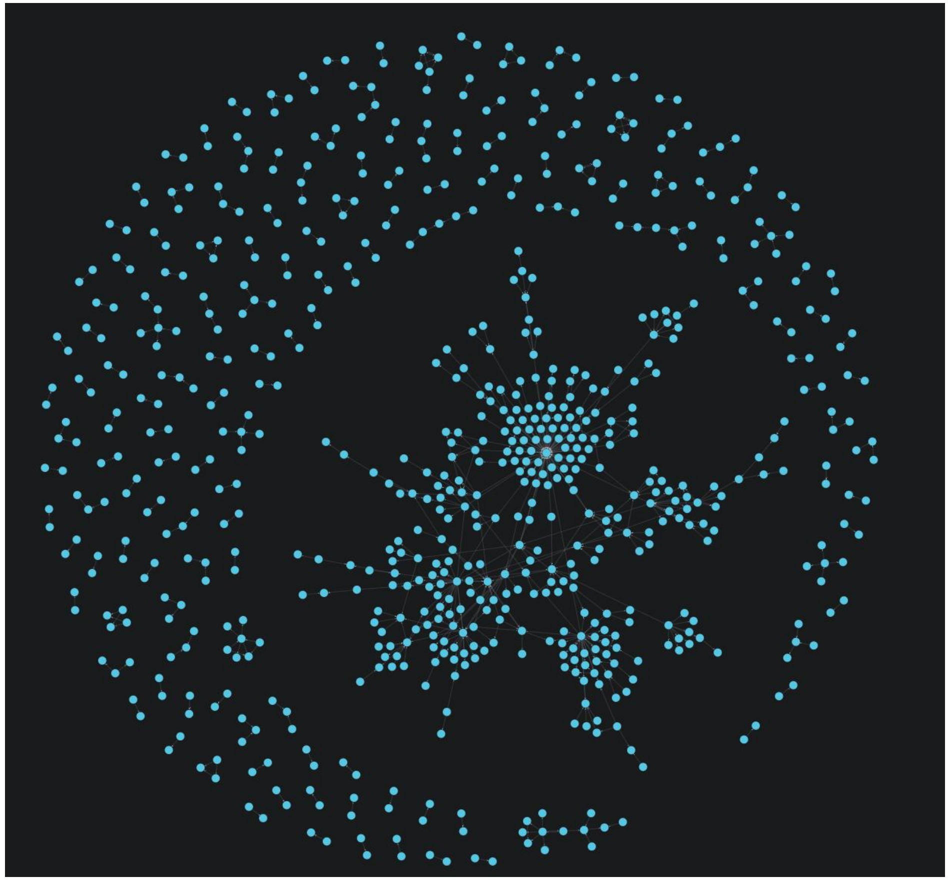
- Count: 887
- Unified entity node type for source & target
- Attributes: name, category, doc_id etc

Relationship

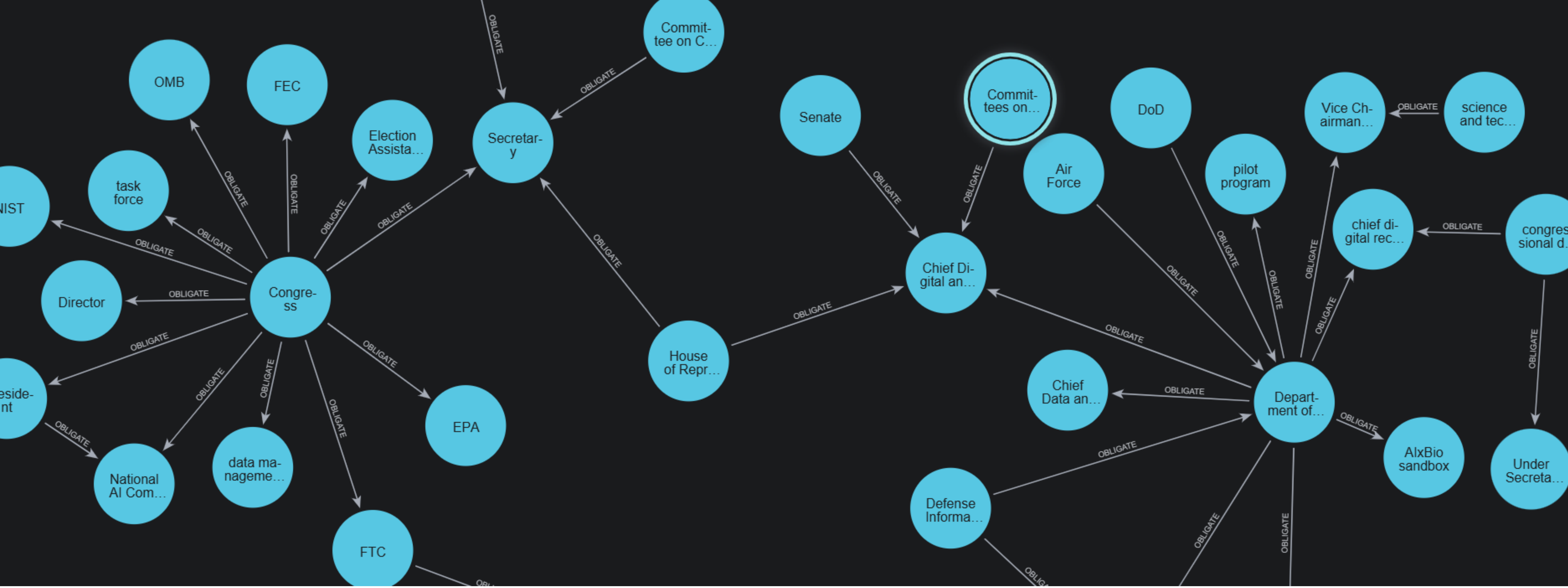
- Count: 653
- Normalized verb property **(DEVELOP / IMPLEMENT / REGULATE / OBLIGATE / REPORT / OTHER)**
- Always directed: actor → target

What This Structure Shows

- Dense central cluster = repeated legal responsibilities
- Sparse periphery = isolated mentions

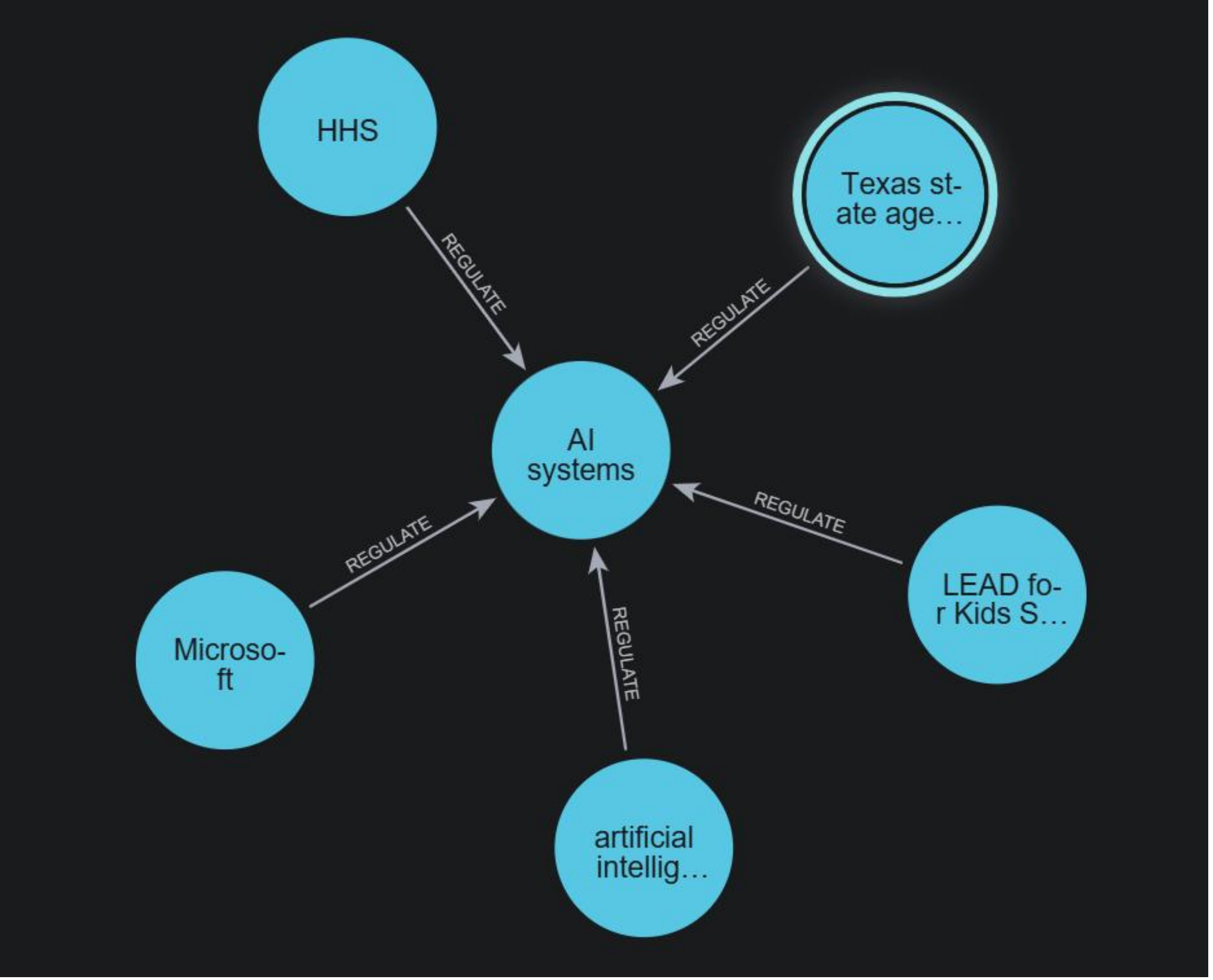


High-connectivity actors cluster in the center; low-connectivity entities form the outer ring.



Obligations cluster around a few central actors

- **Congress is the dominant source of obligations** as it sets requirements for agencies, committees, departments, and program offices.
- **The Department of Defense is the most heavily obligated executive actor,** and it receives many statutory tasks, especially around AI development, governance, data modernization, and oversight structures.
- **State actors (General Assembly, Governor) are also significant obligation sources.** This demonstrates that regulation is not purely federal; states contribute distinct mandates to agencies, councils, and public programs.

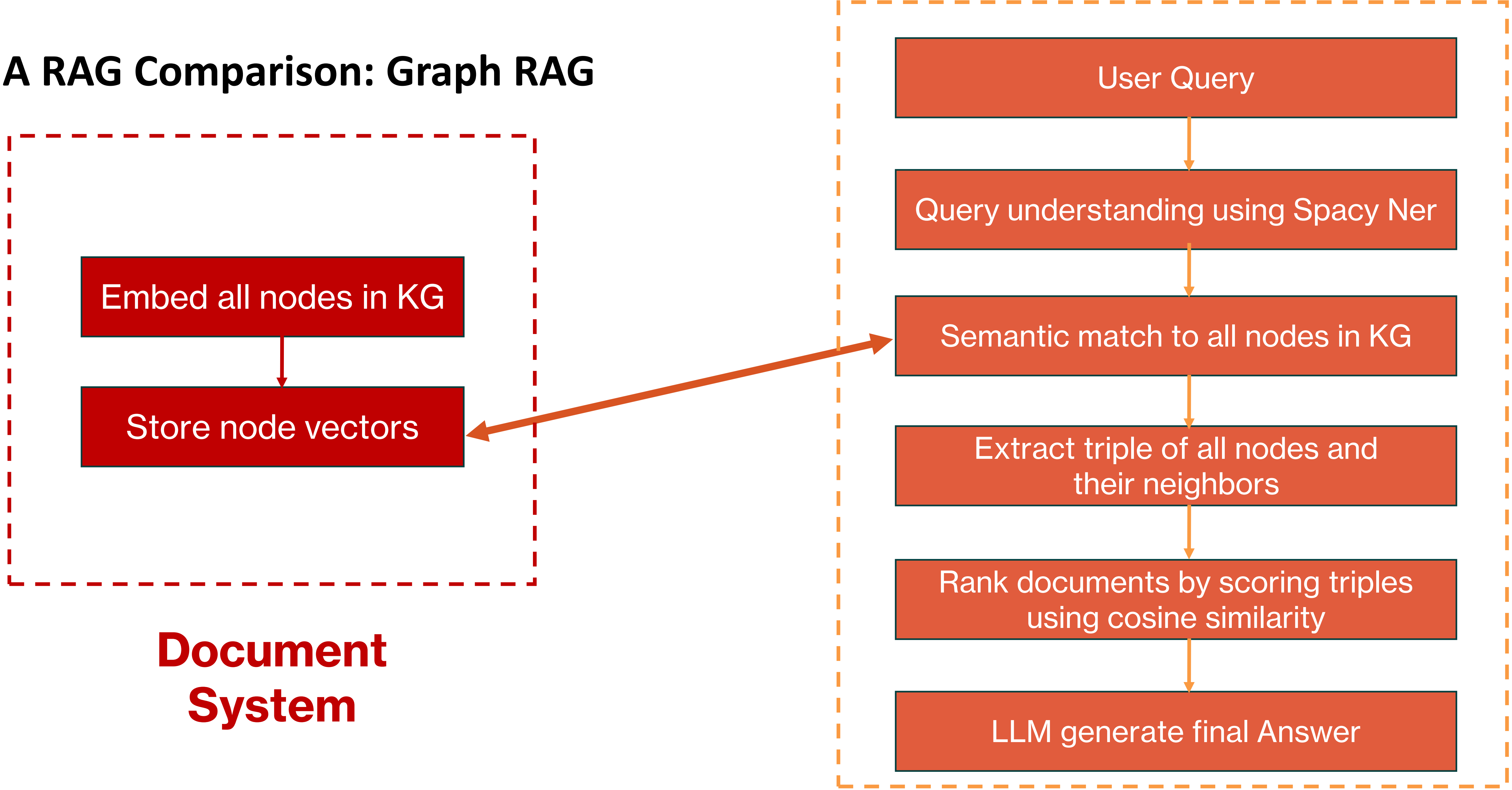


AI regulation appears everywhere, but the KG cannot show what the regulation requires.

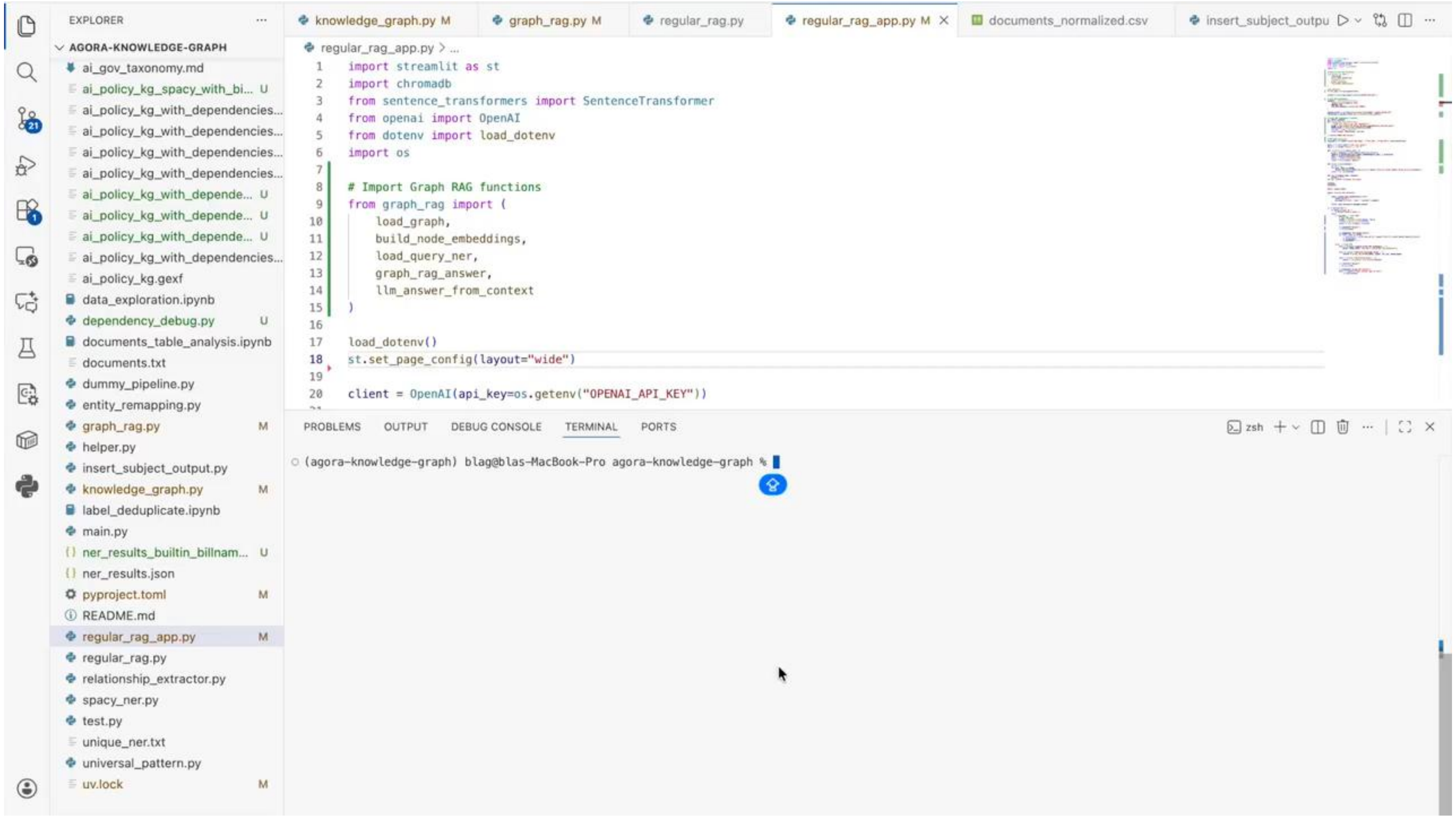
The KG surfaces many **agencies, states, committees, and actors** regulating AI or AI systems. But it cannot represent:

- **the scope of regulation**
- **the content of regulation**
- **the specific harms or risks addressed**
- **exceptions, conditions, or enforcement mechanisms.**

A RAG Comparison: Graph RAG



A RAG Comparison: Streamlit App



A RAG Comparison: Graph RAG

Graph RAG content did pick up keyword, but the triple(s) is meaningless

Graph RAG Context

View context sent to LLM

User Query:
What regulations prohibit use of ai on child sexual abuse image?

Top 3 Most Relevant Documents (Knowledge Graph Triples Only):
==

[doc_662 – Child Exploitation and Artificial Intelligence Expert Commission Act of 2024]
Extracted Triples:
----
prosecution --[OBLIGATE]--> Child Exploitation and Artificial Intelligence Expert Commission Act of 2024

Answer

The Child Exploitation and Artificial Intelligence Expert Commission Act of 2024 obligates prosecution related to the use of AI in the context of child sexua abuse images.

A RAG Comparison: Text RAG

Using iterative chunking of size limit 200, a conventional text retrieval successfully returned highly-relevant results.

Retrieved Chunks

doc_007 – California Senate Bill 1381 (2024) (chunk 1)

Prohibits the production, distribution, or possession of AI-generated depictions involving minors engaging in sexual conduct.

doc_006 – California Assembly Bill 1831 (2024) (chunk 0)

Prohibits the creation, distribution, and possession of child sexual assault material (CSAM) generated by artificial intelligence (AI) in California.

doc_662 – Child Exploitation and Artificial Intelligence Expert Commission Act of 2024 (chunk 1)

and prosecution of child exploitation crimes involving AI.

doc_109 – Protecting Our Children in an AI World Act of 2025 (chunk 0)

Amends title 18 of the United States Code to prohibit child pornography produced using artificial intelligence.

doc_850 – An act creating a temporary state commission to study and investigate how to regulate artificial intelligence, robotics and automation; and providing for the repeal of such provisions upon expiration thereof (chunk 4)

for AI-enabled violations of law and potential restrictions on AI use in weaponry.

A RAG Comparison: Text RAG

Answer

Regulations prohibiting the use of AI in creating, distributing, or possessing child sexual abuse images include:

- **California Senate Bill 1381 (2024):** Prohibits the production, distribution, or possession of AI-generated depictions involving minors engaging in sexual conduct.
- **California Assembly Bill 1831 (2024):** Prohibits the creation, distribution, and possession of child sexual assault material (CSAM) generated by AI in California.
- **Protecting Our Children in an AI World Act of 2025 (federal):** Amends U.S. law to prohibit child pornography produced using artificial intelligence.

Simpler Design Help Scale But Semantic Meaning is Lost

Design Assumptions

- **Triples are Sufficient:** Subject-Verb-Object structures capture enough core signal to be useful.
- **Normalization Works:** Complex legal verbs can be mapped to a finite set of tags (e.g., OBLIGATE).
- **Scale over Depth:** Shallow parsing is acceptable to process massive document volume quickly.

Observed Limitations

- **Loss of Context:** "Orders" vs. "Ordered to" distinctions are often lost in parsing.
- **Complex Predicates:** Multi-word legal terms and nested clauses break simple dependency rules.
- **Semantic Noise:** Structurally valid triples often lack actual legal meaning (e.g., extracting "Section" as a Subject).

THE INHERENT TRADE-OFF

High Scalability & Recall ⇄ **Semantic Fidelity**

MITIGATION: AGGRESSIVE POST-FILTERING

Future Works Focuses on LLM Based Extraction and Better Integration with Graph RAG

Methodology & Extraction

- **LLM-Based Extraction:** Transition to agentic frameworks like LangGraph to better capture semantic nuance.
- **Hybrid Approach:** Use Rule-based methods for structural speed and LLMs specifically for complex/nested clauses.

Evaluation & Application

- **Gold Standard Dataset:** Develop a manually annotated dataset to enable rigorous precision/recall benchmarks (currently lacking).
- **Robust RAG**: Integrate Knowledge Graph implemented in Graph DB with a retrieval pipeline to better support natural language Q&A.

Current: Structural Extraction → Target: Hybrid Semantic Understanding

Appendix

Findings

A. Top obligated actors
B. Obligations towards risk categories
C. Technologies being regulated
D. Reporting requirements
E. Coordination structure
F. Development/implementation network
G. Neighborhood exploration of a specific actor

Learning

Weak verbs and contextual verbs matter because a **legal knowledge graph is not about modeling English sentences**.
It is about modeling **institutional actions, obligations, authorities, and constraints**.
If you treat every verb the text happens to use as an action type, you destroy the abstraction layer that makes a KG useful.

In a KG, the verb must represent a **semantic category**, not a literal word.

- Dependency parsing gives you grammatical structure.

  But legal meaning lives in **semantic roles, legal modality, temporal conditions**, and **domain logic**.
- That's why so little payoff from parsing verbs.

Reading is easy because humans understand purpose, modality, and implication.
Parsing is hard because syntax doesn't map cleanly to meaning.

Meaning comes from relationships, not from entities.

Extraction → Graph construction → Querying → Evaluation

- Accurate entities do **not** guarantee accurate meaning.
- You can extract "Secretary of Commerce" and "personal information" perfectly.
  If the relationship connecting them is shallow, vague, or mis-extracted, the meaning collapses.
- You cannot judge extraction quality in the abstract.
- You only understand whether it worked **after** you put the results into a graph and try to query it.

•**Entities = nouns.**
Nouns don't carry meaning by themselves.
•**Relationships = meaning.**
The verb phrase + structure determines who does what to whom.
•**Triples = the weak approximation of meaning.**
The quality of the triples determines whether the KG becomes intelligent or just a bag of nodes.
•**Legal text is full of multi-word relationships.**
"Shall ensure protection of," "shall be responsible for," "is authorized to establish," "may promulgate regulations for"…
These do not reduce cleanly to a single verb.
•**Entity accuracy without relationship accuracy still produces noise.**
This is exactly what you're seeing.

Ontology

- OBLIGATE – formal “must”, “shall”, duties, mandates
- REGULATE – oversight, enforcement, compliance, governance
- AUTHORIZE – allow, permit, fund, expand power, empower agencies
- EVALUATE – assess, test, examine, audit, analyze
- REPORT – submit information, notify, disclose, file
- SUPPORT – promote, assist, encourage, help
- COORDINATE – collaborate, share, convene jointly
- RESTRICT – prohibit, ban, deny, limit access
- DEVELOP – create, adopt, build, establish, design, improve
- IMPLEMENT – put into practice, carry out, execute
- USE – straightforward operational use

REQUIREMENT
LIFECYCLE STAGE
RISK
DOMAIN
TECHNOLOGY
LEGAL SOURCE
ACTOR